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**MATE ROV 2025**



# **Sunk**

## **ROBOTICS**

**Medford Vocational Technical High School  
Medford, Massachusetts, USA**



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**Andrew Huang - Software Engineer**

**Timothy Hunt - Mechanical Lead**  
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**Benjamin Smith - Mechanical Engineer**  
**Benjamin Wirz - Mechanical Advisor / CSO**  
**Jonas Wirz - Pilot / CEO**  
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Photo 1. Sunk Robotics Team Members, Photo: Noah Lewkowitz

## I. ABSTRACT

**Sunk Robotics** is a unique underwater robotics company specializing in developing cutting edge remotely operated vehicles (ROVs) tasked with performing complex missions in a variety of marine ecosystems. This contributes to the global oceanographic community through the collection of ocean data, servicing of marine infrastructure, and assistance in understanding ocean processes. Sunk Robotics has developed its products in accordance with the 2025 MATE request for proposals to service marine renewable energy, explore shipwrecks, and investigate other underwater debris. Through detailed planning, prototyping, data analysis, and testing, Sunk Robotics has developed its latest generation of products to maximize safety, precision, modularity, and reliability.

**JONA** is a state-of-the-art underwater ROV with eight thrusters, six-axis automatic stabilization, and a modular manipulator system. Controlled with a modular web client, JONA is easily pilottable with any computer and a web browser.

**JENA**, our innovative GO-BGC float, utilizes a custom designed profiling mechanism: a piston-compression buoyancy engine. JENA, featuring an updated sensor package, can easily be profiled to a remotely adjustable depth for data collection.

This document details the design process, budgeting, capabilities, and priorities of Sunk Robotics' 2025 product line.

## II. PROJECT MANAGEMENT

### A. Company Profile

Sunk Robotics, founded in 2021, is based in Medford, Massachusetts, specializing in creating submersible robots to address issues such as climate change, pollution, along with servicing marine renewable energy.

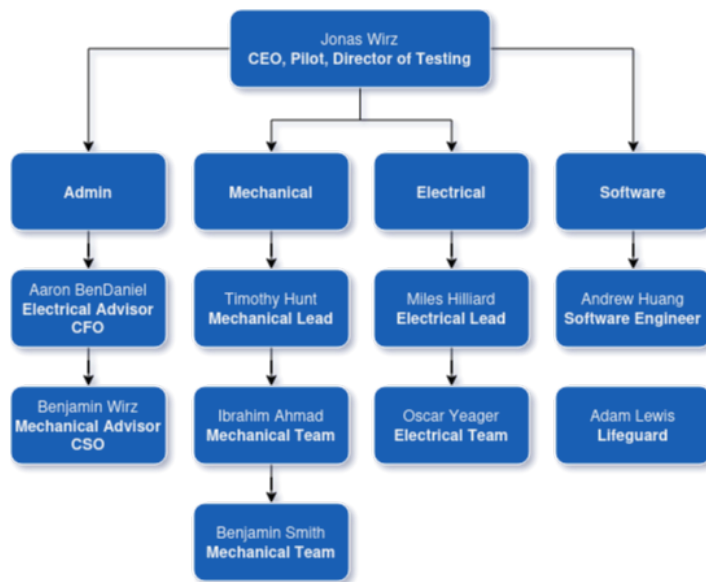


Figure 1. Sunk Robotics Company Organization by Miles Hilliard

Sunk Robotics consists of 10 students from the Medford Technical Vocational High School. While the company is structured hierarchically for technical development, we employ a flat structure for management and team decisions. The company consists of four departments: electrical, mechanical, software, and administration. (see Figure 1) Although the departments usually work independently, all members pitch solutions to problems during daily check-ins.

This year, we felt the effects of ignoring recruitment in prior years. Due to Sunk

Robotics' small size, many of our members have stepped up to take on additional responsibilities.

Our focus this year has been on longevity and reliability. To ensure that the team can prosper in later years, our Lead Float and ROV Hardware Engineers from last year, **Ben Wirz** and **Aaron BenDaniel**, shifted roles to oversee administration, recruitment, and training. Administrative tasks included budgeting, managing workflows, and general documentation. Recruitment has evolved, reaching out to other students both in school and at local sustainability festivals. **Timothy Hunt** assisted with documentation and recruitment.

**Miles Hilliard**, the lead electrical engineer, focused on streamlining the electronic systems on the ROV, such as the main PCBs and the cameras. Hilliard also created various client-based debugging programs to record and analyze practice task runs. Hunt was tasked with creating the mechanical gripper, adapting the manipulator system, creating safer thruster guards, and R&D for hydrodynamic systems. He also acted as the head of community outreach, setting up booths at a sustainability festival. **Jonas Wirz**, along with practicing as the pilot, handled the competition portion and created the props. **Oscar Yeager**, the float electrical engineer, outfitted the Float with a rotary encoder, allowing adjustable remote profiling. **Ibrahim Ahmad**, the float mechanical engineer, outfitted the Float's piston with an updated



motor along with a more reliable power system and an upgraded antenna.

**Benjamin Smith**, a mechanical engineer, did R&D on cable management, along with updating the modular camera housing.

**Andrew Huang**, our main software engineer, spent much of the year learning Sunk Robotics' pre-existing custom software stack. Huang then adapted pre-existing programs to fit this year's call for proposals, along with creating an invasive fish modeling tool.

**Adam Lewis** is our team's official certified lifeguard. Adam ensures safety on the pool deck during our weekly testing.

## B. Company Organization

Sunk Robotics decided to implement project **organization strategies** to keep the team on task and on track to accomplish our goals in a timely manner. In previous years, we utilized Google Sheets to create checklists for departments and team members. This system became increasingly difficult to manage, and we decided to move to a new system.

We opted to use **Notion**, a note-taking application, which allows easy collaboration, tracks changes, and documents work. Using this software has allowed Sunk Robotics to efficiently track and manage all task progress. Sunk Robotics uses various messaging platforms for communication, Google Calendar to track events, and Google Drive for file storage. All code is regularly committed to GitHub, the industry standard in version control software. Fusion 360 and Eagle are

used for all mechanical and electrical design, respectively.

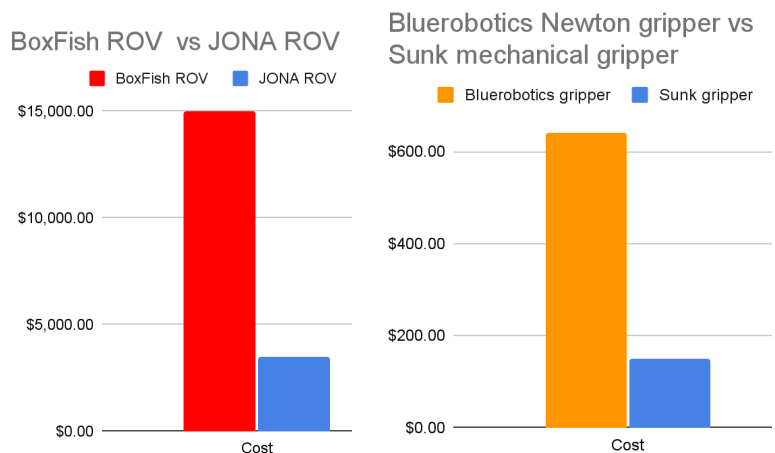
## Team Training

One of Sunk Robotics' major goals for improvement during the 2025 season was **improved team training**. Many important members of the team graduated last year, and this left the team without anyone to fill many important roles, such as lead software engineer and marketing lead.

Sunk Robotics decided to promote several of its younger members to senior positions and challenged them to develop their skills and gain the vital experience that has enabled us to secure a strong future for the team.

New members were tasked with building increasingly complex systems that required them to learn by doing. This hands-on approach to training has had great success, with both new and returning members greatly expanding their knowledge.

## Cost Effectiveness



**Figure 2. Underwater Systems Cost Comparison**  
by Timothy Hunt

Most underwater systems are very expensive. The Boxfish ROV platform costs in the range of **\$15,000**, and the Blue Robotics Newton gripper costs **\$640**. Sunk Robotics has a large advantage in this area. JONA ROV's development cost was only **\$3,500**. Our mechanical gripper package costs under **\$150**. Savings were achieved by creating hardware components and software fully in-house. We built our own software for JONA to enable the usage of Raspberry Pi hardware. Software systems such as ArduSub do not support low cost general-purpose computers such as Raspberry Pi and instead require expensive flight controllers such as a 3DR Pixhawk 1, which we found was priced in excess of \$300.

## Organization Priorities

A main focus of the 2025 season was making sure adequate time and resources were allocated to pilot training. Sunk Robotics created all props from mission tasks the team would be attempting at the 2025 MATE ROV Competition. Ensuring the pilot had sufficient training time was also imperative to our plan. We worked with the Medford Recreation department to schedule pool time.

## Scheduling

The team initially had very little pool access due to priority being given to the school's swim team. After discussions with Medford's athletics director, the team was able to negotiate for enough pool time to allow the pilot to begin their training regimen.

Each weekday, we have a short check-in at 12:20 to help the team understand the state of each project, simultaneously allowing opportunities for suggestions from the team. Most members work on assignments independently from 12:30 to 2:20, as part of the school's robotics vocational shop. This work is usually either hardware assembly, manufacturing, or small scale system testing. Much of the design work is done at members' homes to maximize the time the company can spend manufacturing and testing. The company has pool access from 2:30 to 5:00 every Wednesday. During this time, the company does a mixture of technical testing and piloting practice. In addition to our check-ins and pool time, Sunk Robotics hosts bi-weekly meetings to discuss goals and current tasks.

*Design process continued on [page 7](#).*

# III. DESIGN RATIONALE

## A. Guiding Principles

Sunk Robotics' objective is to create affordable high performance robots to service marine infrastructure and study marine ecosystems. To create the best robots, we follow three overarching design principles.

**Reliability** is our main guiding principle, because a robot that doesn't work is useless. Our robots are designed to minimize all possible points of failure.

**Functionality** is our second guiding principle. We design our robots with the capability to

perform various tasks asked of them, enabling quick and efficient responses to complex situations.

**Modularity** is our third guiding principle, ensuring that our robots can respond to the rapidly changing world of marine technology. Our robots are designed so that new systems can easily be added and old ones can be upgraded.

## Systems Approach

At Sunk Robotics, the main focus of creating a system is solving a problem. This ensures that our ROV is optimized for completing specific tasks while still ensuring that the ROV remains flexible enough to be successful when faced with unforeseen challenges.

Sunk Robotics uses a Pilot-Centric approach to creating and integrating systems. This means that each system is designed around the pilot's plan, instead of the pilot creating their plan around existing systems. Due to the pilot's direct involvement with the tasks, the pilot is the most knowledgeable about what is required for their successful completion.

Our pilot works with all departments to determine what systems need to be created to accomplish mission tasks. The pilot also provides feedback throughout the design and integration process, so that the finalized systems are optimal for completing mission tasks.

## B. Design Process

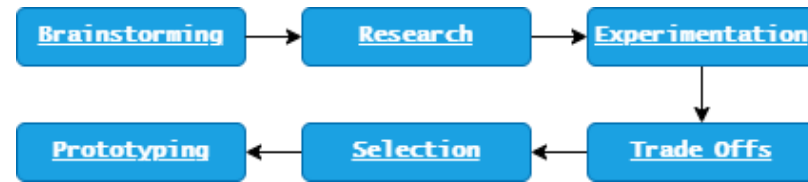


Figure 3. Design Process by Timothy Hunt

### Brainstorming

Every design starts with an idea. Sunk Robotics maximizes the number of ideas it evaluates by using group brainstorming. Once an issue has been identified as worthy of investigation, a brainstorming meeting is called. Each member of the team proposes their unique solutions. After consideration of other members' opinions, the design process proceeds to the next stage.

### Research

Members with relevant expertise research the feasibility of an idea. This research searches for existing products and relevant data, helping with the next stage.

### Experimentation

With information gained from research, a proof of concept is made to conclude whether the idea will function as envisioned. This stage also includes direct data collection and analysis to determine the extent to which the idea functions, assisting with the next stage.



## Trade Offs

An extensive Pros & Cons list is compiled for each idea, and all potential solutions are compared to existing products and systems we have built. This comparison drives the next decision in the process.

## Selection

The whole development team gathers to discuss the options using the Pros & Cons list to select an option of best fit.

## Prototyping

Once an idea has been settled upon, a member with relevant experience is assigned the project to design, prototype, and manufacture, working with others as needed. If prototyping reveals the idea to be inadequate, the selection process begins again.

## C. Design Methodology

Employing the principles of iterative development and testing, a design is prototyped and repeatedly tested. Improvements are integrated into the design, and the cycle is continued until the best possible product has been developed.

## Manufacturing

When possible, Sunk Robotics utilizes two-dimensional design in all of its custom parts. 2D parts are far simpler, cheaper, and faster to manufacture and iterate upon.

When 3D parts are needed, 3D printing is the first option due to cost and efficiency. Though

in cases where higher precision or strength are required, subtractive manufacturing technologies such as CNC turning or milling are employed.

## D. Build, Buy, Reuse

Sustainability is of the utmost importance to Sunk Robotics. As a result, Sunk Robotics values recycling and reusing parts, and a large majority of JONA ROV's parts were made in-house or reused. For 2025, JONA's reused items include: the frame, the thrusters, the electronics tube, the flange mounts, underwater penetrators, various sensors, the tether, most electronic components and cables, along with various pieces of miscellaneous hardware. Sunk Robotics prides itself on custom engineering a vast majority of its electrical and mechanical components, including but not limited to a custom frame, a camera system, circuit boards, connectors, and the tether. Parts are purchased or outsourced only if we don't have the required tools, or would be too unreliable to custom manufacture. For example, Sunk Robotics considered manufacturing our own tube flanges, but it was significantly more expensive to buy the materials to turn our own flanges than it was to buy them pre-made from Blue Robotics. We decided that cost effectiveness was more important than the flexibility of custom tube flanges.

## IV. Testing & Troubleshooting

Testing and troubleshooting are essential aspects of Sunk Robotics' development process. Each system is tested at each stage of development to ensure that minimal time is wasted on non-functional designs. Rapid prototyping is featured heavily at every stage of the development process, and each prototype is tested to ensure functionality. When a system doesn't function as intended, the issue is broken down into pieces that can be independently tested. Each piece of the system is tested in the order of which is most likely to be the issue. After an issue has been identified, we choose between three options on how to fix the issue: purchase a new component, re-manufacture a component, or re-design a component.

### A. Electrical

Electrical difficulties can often be identified by following a standardized process. If an electrical change has been made recently, it is wise to analyze those modifications first. During this time, the adjacent electrical systems are usually broken down and tested individually. This helps to narrow down the possible problematic area, and is especially useful in large scale electrical difficulties that could potentially have various sources.

### B. Mechanical

Mechanical issues typically boil down to three types of problems: breakages, misalignments, and leaks. Breakages are usually fixed by

either changing materials or increasing the amount of material at the broken sections. Misalignments are usually fixed by changing dimensions then re-manufacturing, or adding an adjustment screw. Leaks are tackled by individualizing testing each possible point of failure. After fixing the leak, a vacuum pump is attached to the vent cap, creating a vacuum inside the electronics tube. If after one hour the pressure stays constant, the leak is fixed.

## V. Hardware Overview

The process of designing JONA ROV's hardware was meticulous. 6 goals were set out for the design: neutral buoyancy, 6 degrees of freedom, corrosion resistance, extreme resilience, a modular camera system, and a modular manipulator system.

### A. Frame

Sunk Robotics dedicated substantial time to evaluating various frame designs, considering their suitability for product missions. Each design was carefully compared, weighing its respective advantages and disadvantages, until a single, optimal design was made.

### Buoyancy & Material: Trade Offs

JONA ROV's frame is constructed from 3mm sheet aluminum. This decision involved significant amounts of research and consideration. After calculating the buoyant forces with the Electronics Tube and different frame materials, we found High-Density Polyethylene (HDPE) to be too light,

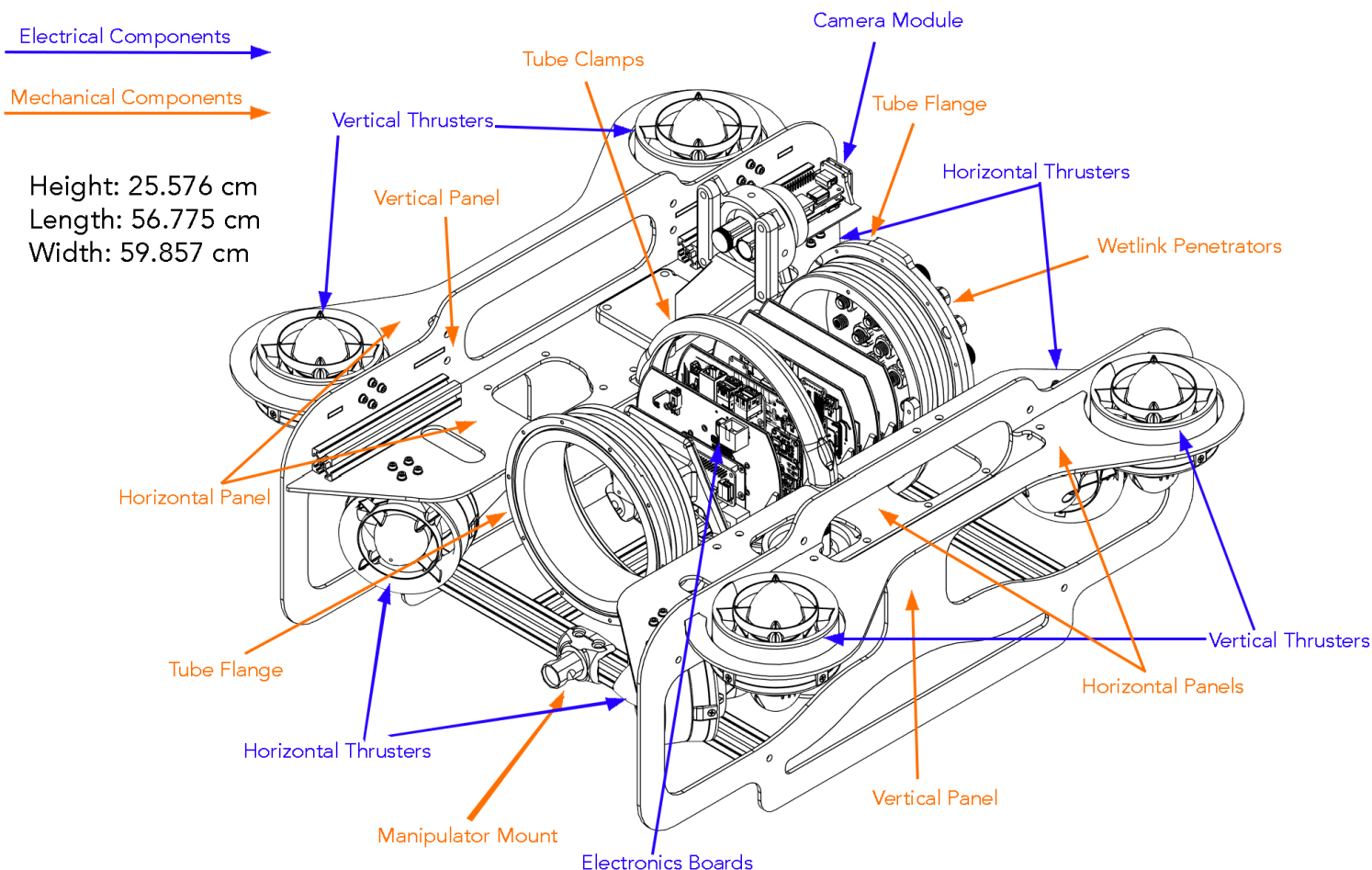


Figure 3. Labeled JONA ROV Overview, by Timothy Hunt

and stainless steel too heavy. Nickel-titanium alloys, often used in ultra high end submersibles, were too expensive. We calculated the frame in aluminum to weigh just enough to cancel out the buoyant forces of the positively buoyant electronics tubes.

## Hydrodynamic Efficiency

After performing a physical and computational study on possible modifications to the ROV to reduce drag force, we discussed how to move forward with those concepts.

In this conversation, incorporating feedback from the pilot, we decided that hydrodynamic efficiency was not worth being the main focus

of this year's design cycle. Although significant performance improvements were shown to be possible, we decided that we were far too deep into the design cycle to make massive overarching changes to the ROV. Another factor driving the team to not focus heavily on optimizing hydrodynamic efficiency was precision. We found increased horizontal drag to allow for more precise movements, improving the pilot's ability to accomplish mission tasks.



## B. Propulsion

The most significant and earliest tradeoff decision was the thruster configuration. Sunk Robotics researched other submersibles for thruster layout inspiration. After compiling the most common and successful options, a meeting was held where the thruster configuration was narrowed down to two options, including a configuration involving a system with 6 motors or one with 8. Other options were determined to be unnecessarily difficult to design a frame and software stack around. During the meeting, the team noted the various advantages and disadvantages of each design.

After discussing the Pros and cons of the thruster configurations, the team with a lot of pilot input decided that mobility and stability were crucial to the ROV's function. The negatives were far less significant than the cons. After analyzing previous years' footage, we realized that a majority of the movement was forward. The mechanical team tested to find that the optimal orientation for the horizontal thrusters was 30 degrees. Sunk Robotics made many similar decisions with our Pro-Con organizer.

| Thruster Configuration      |                             |
|-----------------------------|-----------------------------|
| 6 Thrusters                 | 8 Thrusters                 |
| Pros                        |                             |
| Decreased power consumption | Pitch control               |
|                             | Increased vertical thrust   |
| Cons                        |                             |
| No pitch control            | Increased power consumption |
| Decreased vertical thrust   |                             |

Figure 4. Pro-Con Organizer  
by Matty Harris

## C. Modular Manipulator System

Sunk Robotics realized that due to the complexity involved in servicing marine infrastructure and studying marine ecosystems, no one tool would be sufficient in accomplishing all current and future tasks. That is why JONA utilizes a custom quick-change modular manipulator system. This allows for new tools to be designed, built, and easily implemented to complete any task.

This system is designed for rapid infield tool changes. The octagonal receiver allows for any manipulator to be rotated in 45 degree intervals, further increasing the functionality of each tool. Each manipulator is attached to an octagonal receiver using a neodymium magnet and a retaining pin for added reliability under high load.

### Hook Manipulator



Photo 2. Hook Manipulator  
by Timothy Hunt

The hook is used to accomplish a variety of simple tasks. While mechanical grippers can be more versatile, they are slower to operate,

require more power, are more prone to failure, and are more expensive. When possible, the static hook is. In the unlikely situation that the hook is damaged, it is easy to replace, consisting of 2 water jet aluminium plates and a 3D printed part.

## Mechanical Gripper

The mechanical Manipulator is used when the Hook is insufficient for a task. It consists of a high torque micro servo with custom interchangeable aluminum claws and rubber padding. Sunk Robotics decided that it was far safer to utilize a pre-waterproofed servo. The rubber padding allowed for our innovative low current mechanical manipulator.

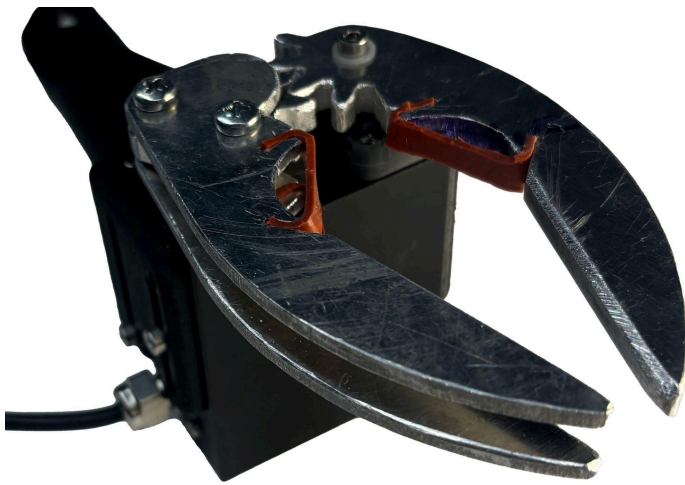


Photo 3. Mechanical Gripper  
by Timothy Hunt

## Software

The mechanical manipulator constantly monitors its current usage. When current spikes above 3 amps, the servo releases by degree until the current drops below the limit. The gripper maintains engagement with the object because of the rubber padding. The padding and the nature of a servo allows a

clamping force to lock the object in place securely. This system allows for a significant reduction in power usage, allowing for greater efficiency when the gripper is engaged.

## D. Tether

### Goals

A high-quality tether is a must for allowing robust communication, power delivery, and maneuverability underwater. The main goals for tether design were to minimize thickness and voltage drop. Reduced thickness lowers cable drag while lower voltage drop allows more available power for the thrusters. Both of these increase the overall performance of the ROV.

### Construction

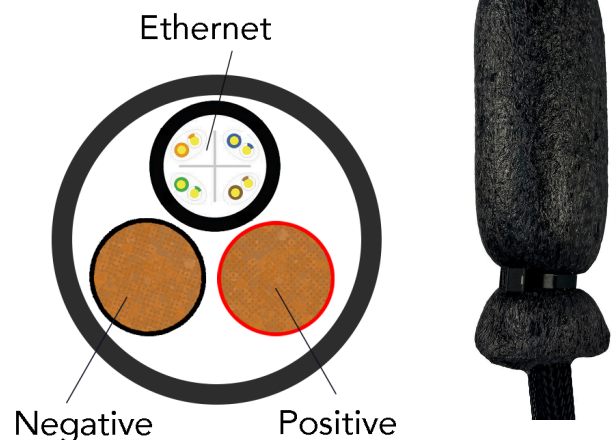
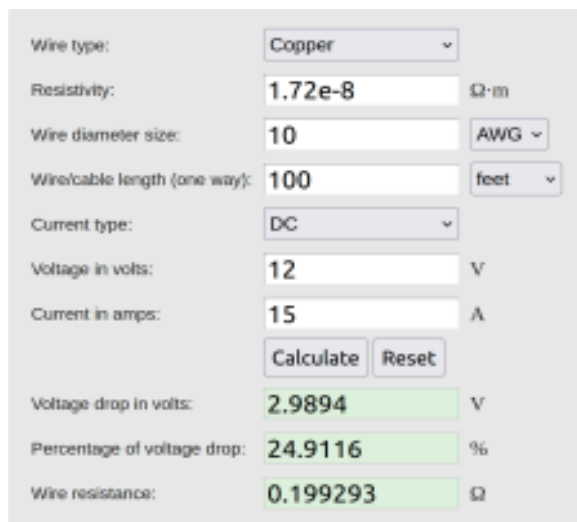


Figure 5. Tether Cross Section & Floatation  
by Timothy Hunt

With standard cable design, these goals are contradictory. In most cases, the thinner the cable, the higher the voltage drop. We minimized this issue by using two 10 AWG ultra high strand count wires for power

delivery and a single CAT 6A Ethernet cable for communication.

All three cables are wrapped in a braided cable sheath. This sheath helps to organize the tether, making it easier to manage.



|                                                   |             |
|---------------------------------------------------|-------------|
| Wire type:                                        | Copper      |
| Resistivity:                                      | 1.72e-8 Ω·m |
| Wire diameter size:                               | 10 AWG      |
| Wire/cable length (one way):                      | 100 feet    |
| Current type:                                     | DC          |
| Voltage in volts:                                 | 12 V        |
| Current in amps:                                  | 15 A        |
| <button>Calculate</button> <button>Reset</button> |             |
| Voltage drop in volts:                            | 2.9894 V    |
| Percentage of voltage drop:                       | 24.9116 %   |
| Wire resistance:                                  | 0.199293 Ω  |

Figure 6. Tether Voltage Drop Calculations by  
Matty Harris

## Innovation

Sunk Robotics made a major innovation for the tether by making it easily detachable, using three Blue Trail Engineering Cobalt Connectors. This modular design allows the tether to be easily disconnected from JONA, making tether management and organization easy and streamlined.

## Buoyancy

The cable's weight causes the tether to be negatively buoyant. This is undesirable as the tether will drag along the bottom of the pool, get caught in props, and pull on JONA. To remove this issue, flotation was added to the tether. Lengths of foam pipe insulation are affixed to the tether at regular intervals. This insulation makes the tether slightly positively

buoyant, allowing the tether to stay out of the way while not pulling up too hard on JONA.

Strain relief is described on [page 22](#).

## E. Electronics Enclosure

One of the goals for JONA's electronic system was to make the design more robust, reliable, and flexible. Employing custom and modularly designed printed circuit boards (PCBs) was the best way to achieve this. JONA's electronics enclosure consists of a single 15 cm diameter clear acrylic tube that houses all drive and control electronics. The clear tube allows for quick diagnosis of electrical issues without exposing the electronics to a possibly hazardous environment. The cylindrical shape of the tube normalizes strain throughout the tube, improving strength.

## Flanges

The ends of the enclosure are sealed with Blue Robotics 15cm Tube Flanges. The hardware team considered manufacturing the flanges in-house with a lathe, but ultimately decided it would not be cost-effective. The cylindrical design also allows for the tube to be sealed using O-Rings rather than gaskets. The advantage of O-Rings is that they can be oriented to form a seal without clamping components together.

## Wet-Link Penetrators

In alignment with Sunk Robotics' dedication to modularity, all of JONA's external wiring goes through removable penetrators to allow for an

easily serviceable system. The alternative to this system would be to directly epoxy the cables through the back plate of the ROV. This would make repairing or replacing a cable significantly more difficult. Sunk Robotics decided that creating custom penetrators would not be worthwhile, as they would be less safe and reliable than professionally manufactured ones.

The rear plate accommodates nineteen M10 penetrators, although only sixteen are currently in use, allowing for future expandability. Eleven Blue Robotics Wetlink Penetrators are used for JONA's eight thrusters, two cameras, and a mechanical manipulator. Three Blue Trail Engineering Cobalt Connectors (two 3-pin power for 12V and GND to the tether and one 8-pin for Ethernet to the tether) are used, along with one Blue Robotics Enclosure Vent and Plug, and one Blue Robotics Depth Sensor.



Figure 7. JONA End Cap Render  
by Aaron BenDaniel

## Innovation: Electronics Skeleton

The O-ring seal allows for one of Sunk Robotics' most useful innovations: a modular, easily removable Electronics Skeleton.

Sunk Robotics' proprietary electronics skeleton consists of one custom manufactured aluminium plate, bolted to the inside of the rear flange. Two custom machined steel bars are threaded into the bottom of the panel. Five resin printed ribs hold the custom designed PCBs. The ribs slide onto the bars and lock in place using two set screws. This mechanism allows for easy adjustment of PCB locations

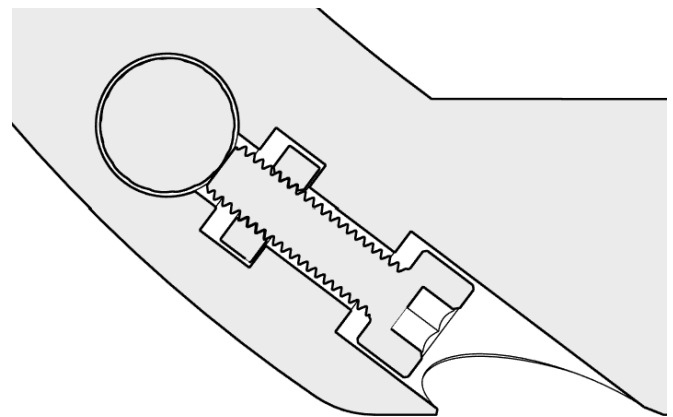


Figure 8. Electronics Skeleton Locking System  
by Timothy Hunt

while maintaining rigidity when locked in place. On the bottom of each rib is a set of mounting points. These allow for custom cable management attachments for different ROV configurations.

In the unlikely case of a breakage, each rib is identical. This makes it possible to carry only one style of replacement for all boards. Each PCB is strategically placed in the frame and is organized based on the specific functions it serves in the ROV. This method of organization not only simplifies the wiring process but also minimizes potential points of failure.

Another main advantage of the Electronics Skeleton is its serviceability. It can be removed



from the tube without any tools. Each board can be independently extracted for troubleshooting or replaced entirely to be upgraded.

## VI. Electronics Overview

### A. Overview

JONA's electronics systems were designed with the goal of maximizing reliability, modularity, and flexibility. The electronics team decided that designing our own custom Printed Circuit Boards (PCBs) would accomplish all these goals.

Splitting the system into 5 PCBs further increases the modularity and serviceability of JONA's electronics, while simultaneously allowing each board to be upgraded without re-manufacturing the whole system. It also enables issues to be troubleshooted by isolating boards and testing them individually. The boards are arranged horizontally so that parts can be easily removed for service or repair.

### Electronics Skeleton

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Figure 9. JONA Electronics Skeleton Render  
by Timothy Hunt

This method of organization not only simplifies the wiring process but also minimizes potential points of failure.

The main advantage of the Electronics Skeleton is its serviceability. It can be removed from the tube without tools. Each board can be independently removed for troubleshooting or replaced entirely to be upgraded.

Each PCB is strategically placed in the frame and is organized based on the specific functions it serves in the ROV. This method of organization not only simplifies the wiring process but also minimizes potential points of failure.

### B. PCB Functions

JONA's control circuitry is composed of five individual PCBs: Logic, Networking, Power Distribution, and two Motor Control Boards

## Power Distribution

Power from the tether first goes to the Power Distribution PCB. This board distributes 12V power to both Motor Boards and has a 12V to 5V converter, which supplies power to all of the logic/control components. A 4-channel ADC is used to measure current and voltage of the 12V and 5V systems.

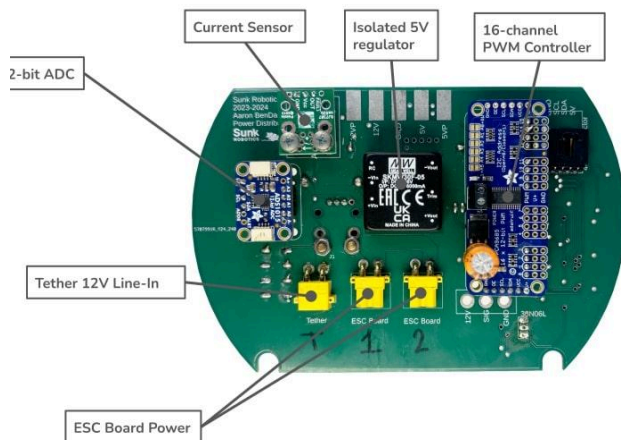


Photo 4. Labeled Power Distribution Board

Photo: Miles Hilliard

## Motor

JONA's Motor Boards each contain four Blue Robotics ESCs along with connectors to manage the power input, signal input, and motor output. These boards take signals from the logic board and convert them into the raw outputs required to control the 8 thrusters.

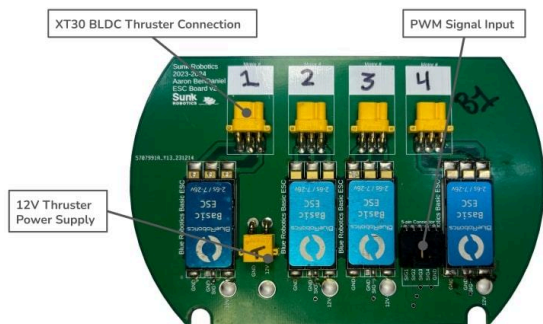


Photo 5. Labeled Motor Board

Photo: Miles Hilliard

## Networking

The Networking Board connects all of the Ethernet-based devices together. It has a power input connection and five RJ45 ports. Three of the ports are wired with 100Mbps Ethernet connectivity along with a custom-built POE protocol supplying 5V power, and the other two are wired with 1Gbps Ethernet connectivity. The board contains a Gigablox Ethernet Switch provided by SwitchBlox with gigabit support to connect all of the Ethernet devices.

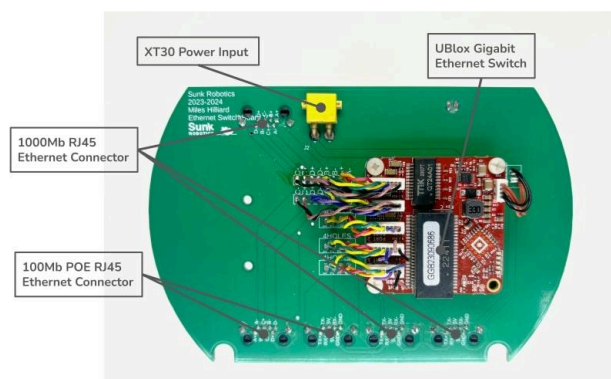


Photo 6. Labeled Networking Board

Photo: Miles Hilliard

## Logic

The Logic PCB contains a 5V power bus, an I<sup>2</sup>C bus, a Raspberry Pi 5, and a 9-axis inertial measurement unit. All of JONA ROV's components are interfaced via either the I<sup>2</sup>C protocol or Ethernet IEEE 802.3.

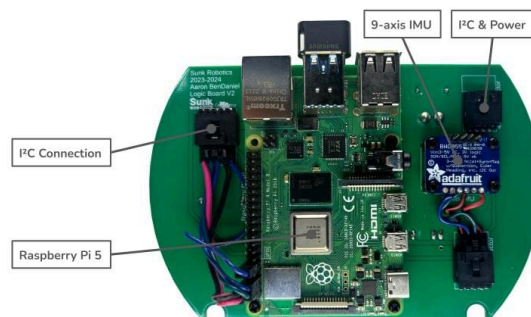


Photo 7. Labeled Logic Board

Photo: Miles Hilliard

## C. Sensor Package

JONA features a minimalistic modular sensor array that is tailored specifically to the requirements of the call-to-action. JONA's sensor package is easily expandable, allowing for future upgrades and modifications as required.

### Pressure

JONA ROV has a Blue Robotics Bar02 pressure and temperature sensor that can calculate depth, which is used for general navigation along with the depth PID system.

### IMU

JONA uses a BNO085 9-axis IMU to sense its orientation in 3D space. This information is used for the pitch, roll, and yaw PIDs, along with enabling the autonomous navigation system.

### Cameras

JONA ROV hosts a custom modular array of camera modules. These cameras are used to allow the pilot to see the surroundings of the ROV with low-latency video feeds. Due to the modular nature of the camera system, additional cameras can be added to suit mission requirements. Each camera has a Raspberry Pi Zero 2 that handles all of the image processing for each camera. Because each camera has independent hardware, the failure of one camera does not mean the failure of the entire ROV's vision system.



Photo 8. Sunk Camera Module  
by Timothy Hunt

## D. Control Station

Sunk Robotics utilizes a single laptop as its main control system. Unlike the typical bulky and complex custom control boxes, our system is highly portable and can run on any personal computer. A simple, familiar, and robust Xbox One controller is used as the main control interface.

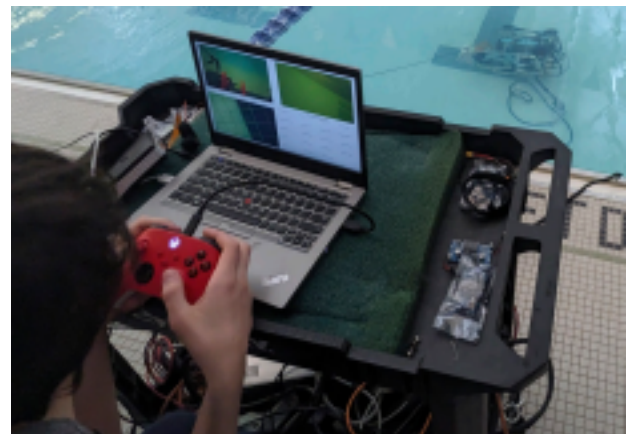


Photo 9. Pilot Practice  
Photo: Aaron BenDaniel

## VII. Software Overview

Software SID found on [page 24](#).

### A. Control Interface

The pilot user interface is built on an intuitive and responsive front end to ensure that even a novice pilot can easily control the ROV. An Xbox One controller is attached to the computer via USB or Bluetooth. JONA automatically communicates with the control device immediately after launch. The frontend client displays low-latency, high frame rate video from each of the ROV's onboard cameras, affording the pilot maximum awareness of their surroundings. The client also displays power usage, orientation, temperature, and depth, allowing the pilot to easily ensure proper and safe operation of the ROV.

To connect, the pilot simply plugs in the tether's Ethernet cable, which automatically establishes a connection to the ROV's server from any web browser. This ensures maximum compatibility and minimizes startup time.

### Rationale

The frontend was built entirely from scratch using Vue.js, a framework for building graphical user interfaces on the web, and TypeScript. The software team developed a custom solution for decoding and displaying video feeds from multiple cameras with

minimal latency, achieving under 170 ms latency. Other libraries and frameworks used include: Tres.js for displaying the 3D model of the ROV and Chart.js for creating graphs of sensor data. The team opted to use Vue.js because it simplified the process of developing a visually appealing, intuitive user interface.

### PID Stabilization

**Stable** and **precise** positioning is imperative to quickly performing mission tasks. Without a stabilization system, the pilot would be occupied by compensating for drift and water currents. To address this, the software team developed an original stabilization algorithm to "anchor" the ROV in place, letting the pilot focus on performing tasks.

There are four independent stabilization systems: depth, pitch, roll, and yaw. Each stabilization system is based on a **Proportional-Integral-Derivative control algorithm (PID)**. A PID controller is a mathematical tool that allows a system to quickly reach and maintain a set position, such as depth or orientation, without overshooting or oscillating. The algorithm calculates the difference between a target value and the actual value, and responds in proportion to this error. How strongly the algorithm reacts to this error depends on the proportional term in the equation. The algorithm also includes an adjustable integral term that sums the error over time, and can help avoid oscillating between large and small error values. The



algorithm also includes an adjustable derivative term that calculates how fast the controller is approaching the set point, used to avoid overshoot. These three terms work together to allow the pilot to set a depth or rotation that will remain constant in spite of water currents.

## B. Video Feed

Our custom made software provides low-latency video feeds from all cameras, ensuring the pilot has maximum situational awareness. While many solutions exist for live-streaming video, few meet the latency requirements for piloting an ROV in real time. As a result, the software team decided to develop its own software. The Raspberry Pi in each camera runs our custom Python script that uses Picamera2, a Python API for interfacing with Raspberry Pi cameras. The software encodes a video stream using MJPEG, a computationally efficient protocol for ultra-low latency video streaming. The software then establishes a WebSocket connection with the top-side client, sending each frame. Although MJPEG requires more bandwidth than other video codecs, it can be encoded and decoded extremely quickly with very little overhead, minimizing overall latency. The software team has found camera latency to be under 170 ms. This is extraordinarily fast and ensures that the pilot can respond to stimuli as quickly as possible.

## C. Mission Tasks

### 3D Scaling

JONA ROV features custom analysis and interfacing software that enables props to be measured accurately, in order to determine their scale. It utilizes images taken directly from any camera on the ROV, using known values to determine unknown lengths. Our system corrects for camera warping, rapidly creating very accurate measurements at variable distances.

### Invasive Carp Modeling

Sunk Robotics has developed an invasive carp modeling software, separate from the rest of the software stack. Like the ROV software, this program can be run from any browser. Data is inserted, and a model is returned, showing an animated map of fish movements throughout the Illinois River.

## VIII. Budget & Accounting

| Income:           |                          |               |                              |                               |             |           |                |
|-------------------|--------------------------|---------------|------------------------------|-------------------------------|-------------|-----------|----------------|
| Type:             | Source:                  | Amount:       | Notes:                       |                               |             |           | Running Total: |
| Corporate Sponsor | Watts Water Technologies | \$ 21,685.20  | Left over from previous year |                               |             |           | \$ 21,685.20   |
| Expenses:         |                          |               |                              |                               |             |           |                |
| Type:             | Date:                    | Company:      | Name:                        | Quantity:                     | Unit Price: | Total:    | Running Total: |
| Purchase          | 10/18/2024               | Adafruit      | Electronic Parts             | -                             | -           | \$ 240.00 | \$ 240.00      |
| Purchase          | 10/29/2024               | Blue Robotics | ESCs                         | 4                             | \$ 38.00    | \$ 158.50 | \$ 398.50      |
| Purchase          | 10/29/2024               | Ebay          | Right angle USB C            | 6                             | \$ 4.39     | \$ 34.48  | \$ 432.98      |
| Purchase          | 10/30/2024               | Ebay          | Brushless Motor              | 1                             | \$ 53.16    | \$ 53.16  | \$ 486.14      |
| Purchase          | 11/6/2024                | Ebay          | Drill Bits                   | 1                             | \$ 21.74    | \$ 21.74  | \$ 507.88      |
| Purchase          | 11/6/2024                | Ebay          | Standoffs                    | 1                             | \$ 32.82    | \$ 32.82  | \$ 540.70      |
| Purchase          | 11/20/2024               | Ebay          | Bolts                        | 1                             | \$ 73.38    | \$ 73.38  | \$ 614.08      |
| Purchase          | 12/5/2024                | Ebay          | Power Supply                 | 1                             | \$ 32.49    | \$ 32.49  | \$ 646.57      |
| Purchase          | 12/13/2024               | Ebay          | Wire                         | 1                             | \$ 98.36    | \$ 98.36  | \$ 744.93      |
| Purchase          | 12/25/2024               | Ebay          | Tachometer                   | 1                             | \$ 117.69   | \$ 117.69 | \$ 862.62      |
| Purchase          | 1/3/2025                 | Ebay          | Cable                        | 1                             | \$ 9.99     | \$ 9.99   | \$ 872.61      |
| Purchase          | 1/14/2025                | Ebay          | Taps                         | 1                             | \$ 32.97    | \$ 32.97  | \$ 905.58      |
| Purchase          | 1/16/2025                | Ebay          | Power bank                   | 1                             | \$ 19.95    | \$ 19.95  | \$ 925.53      |
| Purchase          | 1/22/2025                | Ebay          | Countersink                  | 1                             | \$ 26.99    | \$ 26.99  | \$ 952.52      |
| Purchase          | 1/22/2025                | Ebay          | Tubes                        | 1                             | \$ 21.78    | \$ 21.78  | \$ 974.30      |
| Purchase          | 2/13/2025                | Ebay          | LEDs                         | 1                             | \$ 15.49    | \$ 15.49  | \$ 989.79      |
| Purchase          | 2/24/2025                | Ebay          | Battery Pack                 | 1                             | \$ 8.55     | \$ 8.55   | \$ 998.34      |
| Purchase          | 3/5/2025                 | Ebay          | Shaft Coupler                | 1                             | \$ 15.68    | \$ 15.68  | \$ 1,014.02    |
| Purchase          | 3/26/2025                | ACTIVE        | MATE fees                    | 1                             | \$ 450.00   | \$ 450.00 | \$ 1,464.02    |
|                   |                          |               |                              | Value of Reused Parts         |             |           | \$ 3,573.12    |
|                   |                          |               |                              | Travel Expenses for Regional: |             |           | \$ 65.00       |
|                   |                          |               |                              | Ice Cream (If We Win):        |             |           | \$ 55.00       |
|                   |                          |               |                              | Total Expenses:               |             |           | \$ 1,464.02    |
|                   |                          |               |                              | Total Income:                 |             |           | \$ 21,685.20   |
|                   |                          |               |                              | Remaining Balance:            |             |           | \$ 20,221.18   |

## IX. Safety

Safety is of utmost importance to Sunk Robotics. Along with ensuring the safety of the ROV and those handling it, Sunk Robotics has placed major emphasis on protecting its team members from harm. We accomplished this through a combination of physical systems, deck procedures, and safety guidelines for the development process.

### A. Training

With the guidance of team mentors, Sunk Robotics conducts regular safety training to equip team members with the knowledge to avoid injuries and prevent accidents. Each member is trained in pool safety, electrical safety, tool safety, handling hazardous materials, machine operation protocols, and promoting safe organization efforts. All Sunk Robotics team members must practice basic safety by wearing appropriate clothing, proper PPE, and tied-back hair when applicable. Gloves, masks, ear protection, and safety glasses are required when operating tools and handling certain materials and machines. All team members are trained and supervised by the Robotics instructor and Sunk Robotics mentor, Samuel Christy, when operating potentially dangerous machinery. At Medford Vocational Technical High School, all vocational students must complete an OSHA 10-hour industry course; this means that all members

of Sunk Robotics conduct themselves according to the United States Occupational Safety and Health Administration guidelines.

### B. Mechanical Systems

JONA ROV was designed according to Sunk Robotics and MATE Safety standards. All electrical components of JONA are completely secured and watertight, any potentially sharp edges have been deburred, and strain relief is implemented to reduce the risk of on-deck accidents.

#### Advanced Thruster Guards



Photo 10. Thruster Guards  
Photo: Timothy Hunt

In previous years, Sunk Robotics opted to use ultra-low-profile 3D printed thruster guards. However, despite Sunk Robotics meeting the minimum safety requirements, we struggled with paracord getting caught in the propellers. This year, a new thruster guard design has been implemented on JONA ROV. The new design is built with steel mesh and

completely eliminates the risk of paracord getting caught in the thrusters while still minimizing flow impedance.

## Handles



Photo 11. JONA Handles  
Photo: Miles Hilliard

Sunk Robotics developed innovative clip-on handles to prevent injury when handling the ROV. The handles also allow for greater leverage when maneuvering the ROV for payload changing. Using a custom ridged double taper compliant mechanism, the handles are securely attached to the frame with no flow interrupting external hardware.

## Anti-Crush Gripper System

The mechanical manipulator constantly monitors its current usage. When current spikes above 3 amps the servo gradually releases by individual degrees until the current is below the set software limit. After partially releasing, the gripper is still engaged because of the rubber padding. This system allows for massively reduced power usage, allowing for greater mobility when the gripper is engaged.

## Strain Relief

JONA has a loop based strain relief system located in the rear of the ROV. The system loops the tether cables around a specially implemented metal component attached to the bottom of the frame. The full 360 degree rotation transfers all strain on the tether directly to the frame, preventing damage to the electronic systems.



Photo 12. JONA Strain Relief  
Photo: Miles Hilliard

## C. Deck Procedure

To ensure safety during pool practices, Sunk Robotics team members follow a detailed checklist. When working near water, team members always secure any electrical cords or devices away from moisture. Any member on deck communicates with the ROV pilot and tether manager, and the ROV is thoroughly dried after all operations. The team's laptop and PSU are placed on an elevated platform during all pool operations



to eliminate the risk of electrocution. To maintain safe poolside practice, all members on deck must practice setting up, mission operations, and clean up.

## Team Lifeguard

Sunk Robotics is proud to have incorporated specially trained safety personnel into our company. Adam Lewis is American Red Cross lifeguard and first aid certified; he serves as the company's official lifeguard during all pool training and testing.

## Communication

Communication is a crucial part of Sunk Robotics' safety protocol. The entire pool deck team trains in conjunction with the pilot to ensure smooth task completion. The communication structure, much like most of the team, is centered around the pilot. The pilot works closely with the co-pilot to collect data. The co-pilot gathers data from the control station, maximizing the pilot's time efficiency. The co-pilot then communicates this data to the judges. Also on deck are the tether manager and deck assistant. The tether manager's main role is to ensure the ROV does not become entangled in the tether. When manipulator changes or payload deployment are necessary, the deck assistant prepares the item, and the tether manager deploys the necessary equipment. During the deployment, the deck assistant takes over tether management. When the deck hand is

not needed for payload deployment, they help by organizing the jobsite. Additional jobsite personnel handle JENA Float and do in depth data analysis using software on separate devices. During the entire operation, the pilot and co-pilot direct other personnel through a set of predefined commands, allowing for seamless communications without the possibility of miscommunication.

*Safety checklist found on [page 25](#).*

## A. Acknowledgments

Sunk Robotics is grateful to be fortunate enough to be a part of the MATE program once again. The team expresses its gratitude toward MATE, the Marine Technology Society, MATE II, Mass Maritime Academy, and all the people who continuously work hard to provide this opportunity for students worldwide.

Sunk Robotics thanks the following for their constant support:

- **Watts Water Technologies** for their continued financial support towards the team
- Sunk Robotics mentors **Samuel Christy** and **Noah Lewkowitz**, thank you for your patience, guidance, and time spent fostering an environment where students can thrive. Your dedication to Sunk Robotics means the world to the team.
- **Medford Vocational Technical High School** for supporting Sunk Robotics and allowing the team to utilize school resources and facilities.

- **Shane Colton**, your assistance in educating members of Sunk Robotics in advanced PCB design practices was invaluable.

## B. References

- [datasheets.raspberrypi.com/camera/picamera2-manual.pdf](https://datasheets.raspberrypi.com/camera/picamera2-manual.pdf)
- [www.realpython.com](https://www.realpython.com) - Very helpful Python tutorials
- [www.raspberrypi.com](https://www.raspberrypi.com) - Lots of documentation
- [learn.adafruit.com](https://learn.adafruit.com) - Lots of rich documentation and very helpful tutorials
- [pypi.org/project/websockets](https://pypi.org/project/websockets) - Python WebSockets library
- [github.com/Arvamer/gilrs](https://github.com/Arvamer/gilrs) - Rust game controller library
- Parker O-Ring Handbook - Useful information about using o-rings
- [www.rapidtables.com/calc/wire/voltage-drop-calculator.html](https://www.rapidtables.com/calc/wire/voltage-drop-calculator.html) - Tether Calculations
- [wikipedia.org](https://wikipedia.org) - Source of all good knowledge
- LibreTexts Engineering - PID Controllers - Helpful for understanding how PID controllers work

## XI. Appendices

### A. Control Schema

## B. Software SID

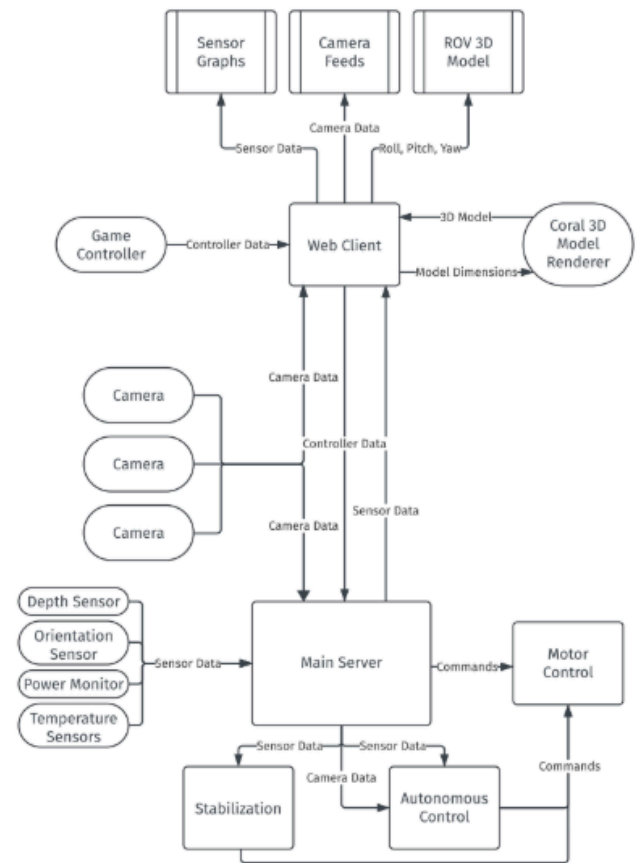


Figure 11. Software SID  
by Matty Harris

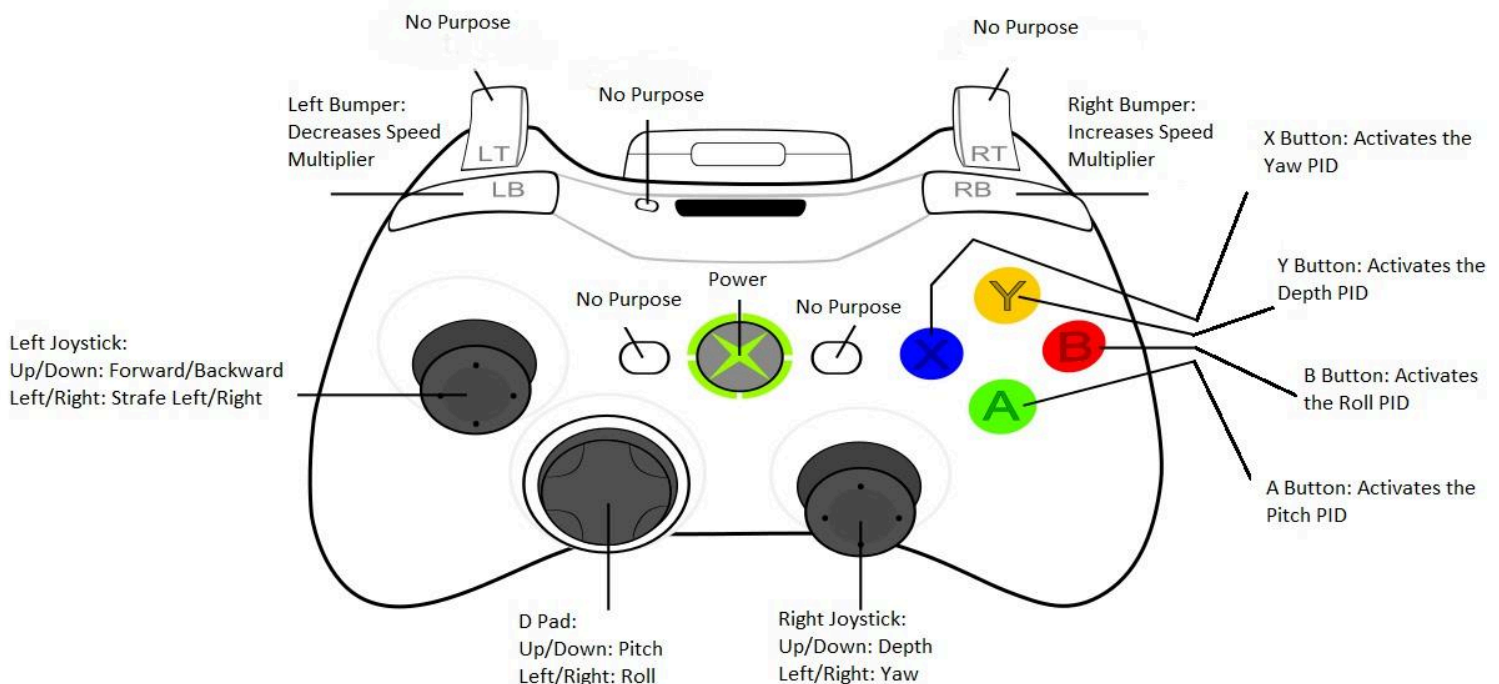
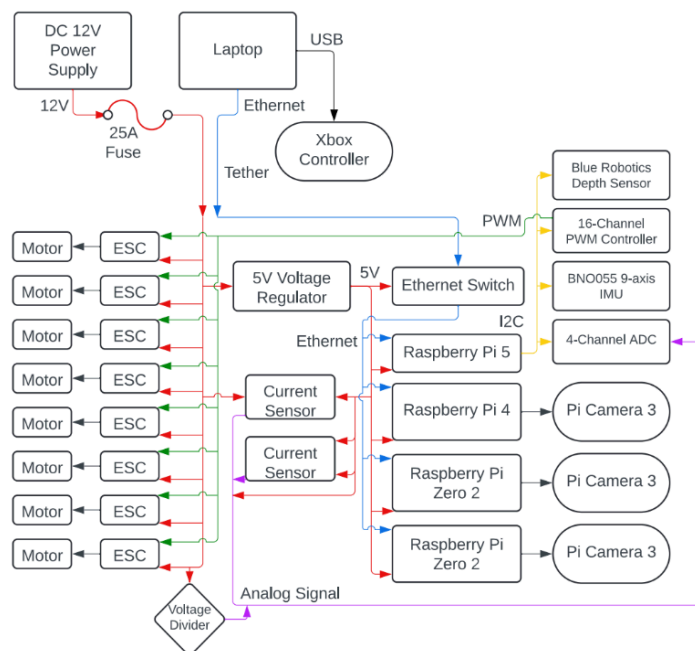


Figure 10. JONA Control Schema  
by Jonas Wirz

## C. Electronics SID



- 8 Blue Robotics T-200 Thrusters - 2.8A each: 22.4A
- 2 Raspberry Pi Zero 2s - 0.35A each: 0.7A
- Raspberry Pi 4: 0.575A
- Raspberry Pi 5: 0.54A
- 4 Low power I2C devices - 0.1A each: 0.4A

**Total Current:  $24.615A \times 150\% = 36.9225A$**

**A 25A fuse will be used**

Figure 12. Electronics SID  
by Aaron BenDaniel

## D. Safety Checklist

### Deck Check:

- ☐ Area is clear of hazards
- ☐ All personnel wearing personal floatation devices
  - Exception for divers
- ☐ ROV is unloaded and ready for power-on
- ☐ Tether is unspooled, untangled, and ready for deployment
- ☐ Props are organized and ready to be given to ROV
- ☐ Tools are organized and ready to be installed

### Inspection:

- ☐ ROV motors are unobstructed
- ☐ Visual inspection of power supply reveals no damage
- ☐ Visual inspection of tether reveals no damage
- ☐ Visual inspection of ROV frame reveals no damage
- ☐ Visual inspection of ROV Electronics reveals no damage
- ☐ Visual inspection of Float reveals no damage
- ☐ All vent plugs are sealed

### Power On:

- ☐ ROV power supply is plugged in
- ☐ Deck attendant announces that power on will begin
- ☐ Anderson Powerpole connectors are mated
- ☐ ROV indicator lights are illuminated
- ☐ ESC startup sound heard
- ☐ Webclient connected
- ☐ Camera feeds started
- ☐ Motor control started

### ROV Launch:

- ☐ Deck attendants confirm ROV is ready for launch
- ☐ Pilot requests ROV launch
- ☐ ROV is lowered into the simulated moon pool
  - Deck attendants hold ROV in place
- ☐ Inspect all watertight enclosures for leakage
  - If leakage is detected, follow leakage procedures
- ☐ Pilot requests ROV release
- ☐ Deck attendants release ROV

### Operational duties:

- ☐ Deck attendants manage tether
- ☐ Deck attendants retrieve objects from the ROV at the discretion of the Pilot
- ☐ Deck attendants give objects to ROV at the discretion of the Pilot

### ROV Retrieval:

- ☐ Pilot bring ROV to simulated moon pool
- ☐ Pilot requests ROV removal
- ☐ Deck attendants remove ROV from water and set on Deck
- ☐ Pilot confirms ready for shutdown
- ☐ Anderson Powerpole connectors unmated
- ☐ Power supply unplugged

### Leakage Detected:

- ☐ Immediately unplug Anderson Powerpole connectors
- ☐ Remove ROV from water
- ☐ Remove Main Tube vent plug
- ☐ Remove electronics frame
- ☐ Drain water from tube
- ☐ Testing concluded for the day
  - Main Tube given at minimum 6 hours to air-dry

Figure 13. Safety Checklist  
by Zach Bertocchi